

High-Throughput Biochemistry: Methods and Motivations for Measuring Equilibrium and Rate Constants at Scale

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Abstract

While algorithms now regularly predict accurate static protein structures from sequence, accurate and quantitative prediction of protein functions remains an unsolved challenge. Progress toward this goal will require generating new data to train these models. These data—measured at the scale and consistency required for functional prediction—do not yet exist, in part because there is no consensus on which data best support prediction or how to collect them. Here, we argue that achieving general functional prediction starts with building a “PDB of function”: a large, standardized repository of equilibrium and rate constants that describe protein function. These constants reflect a unified representation of energetic effects on a common scale across proteins and functions, allowing extrapolation to unmeasured sequences and conditions. Assembling this repository will require high-throughput assays that measure constants for many sequence variants in parallel under identical conditions, circumventing inconsistencies that limit existing databases. We: (1) provide a primer on how to construct physical models and interpret equilibrium and rate constants, (2) survey the current landscape of available measurements, and then (3) discuss emerging high-throughput methods capable of measuring binding, conformational stability, and catalysis for thousands to millions of variants. Finally, we conclude by proposing what a community-scale effort for assembling such a repository would require.

Keywords:

Function, Binding, Catalysis, Stability, Ensemble, High-Throughput

Introduction

The ability to accurately and quantitatively predict how a protein’s sequence encodes its function remains a foundational goal of biochemistry^{1–3}. Such an ability would reveal the direct consequences of sequence variation (i.e., anticipating which mutations cause disease, which antibodies neutralize a pathogen, or how a regulatory variant alters gene expression) and enable design of enzymes, binders, and sensors without any need for screening. However, the relationship between protein sequence and function is extraordinarily complex, both in the vastness of potential sequence space and the diversity of functions that proteins perform. Generalizable function prediction from sequence has not yet been achieved; although there has been tremendous progress in some areas (e.g., predicting certain classes of protein–protein interactions)^{4–6}, the ability to extrapolate to diverse proteins and functions or accurately predict interaction strengths and rates remains unrealized^{7,8}. Meaningful progress toward this capability will require generating data that: (1) quantifies how sequence changes impact multiple functionally relevant properties at massive scale, (2) uses a universal scale that allows data across different proteins and assays to be combined and compared, and (3) enables extrapolation to unmeasured sequences and conditions.

These requirements would be met by a standardized, physically grounded, and open-source data repository. The Protein Data Bank (PDB) provides a powerful example: algorithms now predict high-accuracy structures of previously uncharacterized protein sequences because they were trained on hundreds of thousands of high-quality structures deposited by many researchers in the PDB over the last 50 years^{1,4,5,9–11}. Despite the sheer vastness of sequence space, structure prediction is possible because proteins are built from a limited and recurring set of physicochemical components and their interactions. The same interactions that determine protein structure also underlie functions, such as binding and enzymatic catalysis, suggesting that

function should also be amenable to prediction given the right data. Yet, these efforts are currently bottlenecked by the fact that quantitative functional data at the necessary scale do not yet exist.

In this review, we argue that collecting equilibrium and rate constants—which quantify populations of molecular states and the rates of interconversion between them—with high-throughput experiments is essential for progress in predicting protein function. These constants describe molecular processes as diverse as binding, folding, and catalysis on a universal scale. High-throughput experiments make it possible to measure these quantities across broad regions of sequence space under consistent conditions, making measurements directly comparable. Together, these approaches will allow us to build databases of physically grounded measurements and couple them with modern computational models to advance the goal of generalizable function prediction.

We begin with a primer on the measurement and interpretation of equilibrium and rate constants associated with protein function. We then discuss three classes of protein behavior—binding (to DNA or other proteins), folding and sampling conformational ensembles, and enzyme catalysis—considering for each the relevant physical models, how the constants can be measured, and the current technologies capable of measuring them at scale.

1. Why Measure Equilibrium and Rate Constants?

Despite substantial diversity, all biomolecular functions—such as binding, folding, and catalysis—are governed by a common set of physical rules. Equilibrium (K s) and rate constants (k s) describe fundamental biomolecular behaviors that drive macroscopic phenotypes and provide a universal scale to interpret and evaluate functional changes. These underlying physical determinants are one way to define protein function, but it may also be defined as the emergent and observable, higher-level phenotype a protein exhibits under a given set of conditions. Phenotypic assays are often cheaper, easier to scale, and sometimes the only accessible (or directly relevant) measurement for the problem at hand. Predicting within that phenotype may not require resolving the underlying physical determinants, but the ability to extrapolate beyond the conditions and system in which they were measured will be limited. In contrast, assays that return equilibrium and rate constants are more demanding in design, execution, and interpretation, but provide unique opportunities for mechanistic interpretation and, perhaps more importantly, extrapolation to unstudied systems and conditions. This section serves as a primer on the meaning of equilibrium and rate constants before returning to why their properties uniquely enable generalizable function prediction.

To describe protein function in a unified way, this section establishes the general model in which a given (bio)molecule occupies different states within a conformational ensemble (**Section 1.1**). The distribution of these states and the rates at which they interconvert, in turn, are governed by their statistical probabilities (**Section 1.2**). These probabilities shift dynamically in response to changes in the environment (such as unfolding with temperature, or binding with ligand concentration), leading to observable functional behaviors. When these probabilities are mapped onto a universal scale—energetic differences, with standard reference points—we obtain a unified description of most biomolecular behavior (**Section 1.3**). Ultimately, these probabilities and energies are only useful if they can be robustly quantified, which is achieved through the systematic measurement of equilibrium and rate constants under known conditions (**Section 1.4**).

Finally, we interpret these measurements using free energy diagrams (**Section 1.5**) and consider how mutations alter protein function (**Section 2**).

We explicitly note that equilibrium and rate constants measured in biochemical assays are “apparent” values (K_{app} and k_{app} or K' and k') dictated by specific experimental decisions. However, there are established guidelines for selecting these conditions^{12,13}, and we can reliably extract underlying physical relationships from these measurements by systematically accounting for environmental effects (e.g., pH, ionic strength). We discuss this further in **Section 1.4c**.

1.1. Defining States.

The first step in collecting physically grounded data is to construct an accurate physical model of the system being investigated. The model must do two things: (1) define observable and distinguishable states; and (2) enumerate ways in which the states can interconvert. Every model is a simplification: a defined state is almost always a macrostate, representing an ensemble of multiple underlying microstates that each contribute to the observed state. Defined states should collectively account for all accessible microstates in the system, which establishes conservation of mass. The model should invoke the minimum number of states required to explain a given set of experimental observations and be updated as new experimental approaches reveal new states^{14–17}.

Throughout this section we will use a simple molecular behavior—*isomerization*—as our guiding example (**Figure 1**), as the behavior of simple molecules can be modeled in largely the same way as large and complex biomolecules and help explain protein-level observations^{18–21}. In the *isomerization* of a peptide bond, we can define two fully distinguishable states—*trans* and *cis*—with a single pathway between them (**Figure 1a**). We need not model the underlying microstates (i.e., all possible rotamer states of every bond) explicitly, because our measured ratio of states integrates over them. Molecular changes which effect these microstates will correspondingly alter the measured ratio.

Describing how two states interconvert requires invoking a transition state, a fleeting conformation that is accessed as one state transitions to another. Transition states cannot usually be observed directly, but they govern the rate of conversion, making them essential for modeling and understanding how biomolecular processes occur.

Equilibrium is a natural reference point for interpreting the system. As discussed in the following sections, at equilibrium the populations of states are set by their free-energy differences. Practically, this ensures the measurement will be constant over time, and it defines how a system will tend to evolve when not at equilibrium. The latter point is particularly vital for biology, which is rarely at equilibrium.

1.2. Probability: A Unifying Description of (Bio)molecular Behavior.

Once we have defined states, we can investigate their differences using the language of probability, which quantifies fractional occupancy of each state. In the case of peptide bond *isomerization*, the probability of the *trans* state of the model peptide bond *N*-methylacetamide is ~99% at equilibrium, whereas the probability of the *cis* state is ~1% (**Figure 1a**)¹⁸. Modifications such as altering the functional group identities of adjacent amino acids can shift these probabilities. Because the distribution of states is observable, the effects can be quantified (**Figure 1b**).

Probability also defines transitions from one state to another, quantifying how likely a transition is over time; the relative probabilities of entering a state versus leaving it determine the equilibrium abundance of a state²². Single-molecule studies that track stochastic transitions over time can thus measure both the probabilities of occupying different states (the fraction of time spent in each state) (**Figure 1c**) and the rates of transition between them (the distribution of the dwell times within a state) (**Figure 1d**).

Perturbing environmental conditions shifts these probabilities. Increasing temperature increases the rates of chemical reactions by increasing the probability of reaching the transition state (which involves large, improbable changes in chemical bonds). Adding a denaturant like urea increases the probability of a protein's unfolded state relative to its folded state (**Section 5**). Finally, changing ligand concentration modifies the probability of a protein occupying the bound state (**Section 4**).

1.3. The Boltzmann Distribution: From Probabilities to Energies.

While understanding the impacts of most environmental conditions on the probabilities of states requires direct systematic variation via experiment, impacts due to temperature are accounted for by the Boltzmann distribution (provided that temperature changes do not fundamentally change the number or identity of previously defined states). The Boltzmann distribution connects probabilities to a new quantity—energy—via the following relationship (**Eqn. 1**):

$$\text{Eqn. 1: } P \propto e^{\frac{-E}{k_B T}},$$

where P is a probability, E is energy, k_B is the Boltzmann constant, and T is temperature. The relationship conveys two core concepts: (1) the probability of a state decreases exponentially as its energy increases; and (2) the extent of this difference is scaled by the thermal energy $k_B T$.

The ratio of probabilities of two states depends on the difference in energy between them (ΔE) by **Eqn. 2**:

$$\text{Eqn. 2: } \frac{P_2}{P_1} = \frac{e^{\frac{-E_2}{k_B T}}}{e^{\frac{-E_1}{k_B T}}} = e^{\frac{-(E_2 - E_1)}{k_B T}} = e^{\frac{-\Delta E}{k_B T}}.$$

Two states with a 10-fold difference in probability possess an energy gap of $\ln(10) k_B T$, or 1.36 kcal/mol at room temperature (25 °C, 298 K); a 100-fold difference doubles this gap to 2.73 kcal/mol. Although $k_B T$ evaluates to units of energy, it is often converted to units of energy per mole by multiplying by Avogadro's number (N_A); formally, this shifts the system to units of RT , where R is the ideal gas constant ($R = N_A k_B$)^{23,24}. For the rest of this review, we will discuss systems of molecules and use RT and kcal/mol for consistency.

This specific form of the energy term ΔE depends on the constraints of the system. For biology and biochemistry—which typically occur at constant temperature and pressure—Gibbs free energy (ΔG) is customary, as it accounts for changes in both internal energy (enthalpy) and molecular freedom (entropy) within defined biological systems²⁵. This ΔG specifically refers to the statistical mechanical difference in the Gibbs free energy between two states; as described below, measuring these free energy differences allows us to quantify the effects of changes to biomolecular sequence or the environment on distributions of states and transitions between them.

1.4. *Ks and ks: Our Measurable Quantities.*

Free energy differences are quantified primarily by measuring (apparent) equilibrium (**Ks**) and rate constants (**ks**) using techniques described in **Sections 3–6**. As stated in **Section 1.2**, the rates of transitions are what define the equilibrium state; however, we discuss equilibrium constants first because they are typically easier to measure.

1.4a. *Equilibrium Constants.*

For two interconverting states S_1 and S_2 at equilibrium, **K** is defined by the ratio of S_2 to S_1 (**Figure 1a**). This equals the ratio of probabilities given in **Eqn. 2**, connecting **K** to an energy difference (**Eqns. 3 and 4**):

$$\text{Eqn. 3: } \frac{P_2}{P_1} \rightarrow \frac{[S_2]}{[S_1]} = K = e^{\frac{-\Delta G}{RT}};$$

$$\text{Eqn. 4: } K = e^{\frac{-\Delta G}{RT}} \rightarrow \Delta G = -RT \ln(K).$$

For unimolecular processes, **K** is dimensionless and can be evaluated directly within the logarithm (**Figure 1a,b**). If **K** is defined for a bimolecular process (e.g., binding, **K_D**, which carries units of concentration; see **Eqn. 10** and **Figure 3** in **Section 4**), then a standard reference point becomes necessary. For units of concentration, 1 M is chosen as the standard state, but this does not mean that 1 M ligand (or protein) was required in the experiment.

1.4b. *Rate Constants.*

As molecules undergo thermal fluctuations, they will eventually sample high-energy, low-probability configurations that allow them to convert from one state to another. Transition-state theory²⁶ holds that a low-energy ground state is in a quasi-equilibrium with its transition state (or, more formally, an activated complex near the transition-state barrier, **Eqn. 5**):

$$\text{Eqn. 5: } K^\ddagger = e^{\frac{-\Delta G^\ddagger}{RT}},$$

where $K^\ddagger$ is the quasi-equilibrium constant and $\Delta G^\ddagger$ is the activation free energy. To convert this equilibrium relationship to a kinetic rate, the Eyring-Polanyi equation introduces a scaling factor—the universal frequency factor $k_B T/h$, where h is Planck's constant—that dictates the frequency (units of per second) at which the transition-state barrier is crossed (**Eqn. 6**):

$$\text{Eqn. 6: } k = \frac{k_B T}{h} e^{\frac{-\Delta G^\ddagger}{RT}}.$$

(Note that the full equation formally includes a transmission coefficient, κ , but this is commonly assumed to be unity and not included.) The observed rate (or velocity, v) of a reaction is the instantaneous change in concentration with time. For a unimolecular reaction (or other first-order reaction), the velocity is the product of the rate constant (**k**) and the concentration of the relevant state, $v = k[S]$, with units of M/s. The velocity of a reaction therefore diminishes as the concentration of that state decreases.

Systems reach equilibrium when the velocities of the forward and reverse reactions are equal, or (**Eqn. 7**):

$$\text{Eqn. 7: } k_f[S_1] = k_r[S_2],$$

where k_f and k_r represent rate constants for the forward and reverse reactions, respectively. Detailed balance requires the ratio of these rate constants to equal K at equilibrium, providing another way to represent this state function (**Eqn. 8**):

$$\text{Eqn. 8: } \frac{k_f}{k_r} = \frac{[S_2]}{[S_1]} = K$$

A common non-equilibrium condition is “steady state”, in which the concentrations of the states of the molecule of interest remain constant, but forward and reverse reaction velocities are not equal such that there is flux through the system. (This is especially relevant to enzyme kinetics, discussed in **Section 6.1**.)

1.4c. On the Condition Dependency of Apparent K s and k s in Biochemical Assays.

Equilibrium and rate constants are modulated by environmental conditions such as temperature, pH, ionic strength, etc. Consequently, equilibrium and rate constants measured under different conditions are not necessarily directly comparable, even though each constant is accurate for the specific conditions under which it was measured.

There are two primary ways to account for this variability. The first is to report constants with all experimental/environmental conditions comprehensively annotated and/or standardized^{12,13}, allowing one to account for the variability arising from differences in conditions. The second is to determine $\Delta\Delta G$ s (or $\Delta\Delta G^\ddagger$ s) from the ratio of K s (or k s) relative to a common reference protein under identical environmental conditions (for example, by comparing mutants to a wild-type construct, **Eqn. 9**):

$$\text{Eqn. 9: } \Delta\Delta G = -RT \ln \left(\frac{K_{mut}}{K_{WT}} \right).$$

Expressing effects as $\Delta\Delta G$ relative to a common reference removes any contribution to the free energy that is shared between proteins (for example condition-specific effects that perturb both equally).

Ultimately, the ideal experiment systematically varies both biomolecular sequence and environmental conditions, which allows building more detailed physical models. As one example, changing pH alters the protonation state of ionizable groups according to their pK_{aS} , which represent a $-\log_{10}K$ for ionization states. If one or more of these groups impact the energies of one conformational state or the other (e.g., folded and unfolded), changing pH will alter the relative probabilities of these states and thus the measured K or k s for that reaction. This is especially common in pH-rate profiles of enzymes, where the reaction equation is explicitly modified to include this new state. As another example, changing salt concentrations can alter electrostatic interactions essential for binding or catalysis. In both cases, these effects can be measured and quantitatively accounted for by determining equilibrium and rate constants across multiple environmental conditions^{27–31}.

1.5. Free Energy Diagrams.

The K s and k s of a system can be represented using a free energy diagram (**Figure 1e**). These diagrams assign a free energy to each state along a given reaction coordinate (e.g., bond rotation, folding pathway, etc.); the heights of the barriers between states provide a visual representation of the rate constants associated with each transition. More complex representations include three-

dimensional (or higher dimensionality) landscapes³², but even simple 2-D free energy diagrams provide a powerful tool to describe biomolecular events.

The next section uses free energy diagrams to examine the effects of mutations. For **Sections 4–6**, we provide a physical model and reference free energy diagrams for each of three canonical protein functions (binding, folding stability, and catalysis) and summarize current progress in measuring physical constants.

2. The Value of Physically Grounded Protein Functional Data.

Varying protein sequence and measuring equilibrium and rate constants allows us to explain how sequence dictates function by explicitly quantifying underlying physical interactions. Despite substantial diversity in function, impacts of all perturbations can be measured as changes in free energies (ΔG s).

The most common experiment probes sequence–function relationships by mutating a single residue and measuring the functional effects. To test if two residues are functionally coupled, we can perform a double-mutant cycle (**Figure 2a**) in which we quantify the effect of mutations to both residues alone and in combination^{33,34}. If the residues act completely independently, the effect of the double mutant should equal the sum of each mutant individually. If the residues are functionally coupled, the effect of the double mutant should deviate by a residual ($\Delta\Delta\Delta G$) that quantifies the amount of coupling (**Figure 2b,c**). These residuals enable mechanistic interpretation (i.e., the assignment of a direct physical interaction or the invocation of an unmodeled biochemical state, **Figure 2d–f**). This iteration between creating a physical model, comparing the predictions from that model to observations, and leveraging the disagreement to update that model, is generalizable and provides a scalable architecture to combine individual processes and predict general protein behavior, multi-step processes, and even emergent phenotypes like cell growth rates in virtual cells^{35–38}. Each time this approach fails to be predictive, the discrepancy itself is informative, demonstrating that the existing model is incomplete, and providing a quantitative benchmark to evaluate new models^{39–41}.

Taken together, the properties provided by physically grounded measurements—a common scale, a defined null expectation, and diagnostic residuals—are what make equilibrium and rate constants the data required to realize generalizable function prediction. If the energetics associated with fundamental biomolecular behaviors can be learned and modeled, then problems that appear extrapolative (whether to new sequences, new properties, or new conditions) become interpolative within a single physical space. This will not be achieved by simply collecting more data; instead, building predictive models will require measuring equilibrium and rate constants at scale and under defined and systematically varied conditions.

A growing set of technologies is making the measurement of equilibrium and rate constants tractable at scale. We turn next to what such data currently exist and needs for future data collection.

3. Databases of Constants: Current Efforts and Considerations.

Current databases of equilibrium and rate constants primarily include data from three sources: (1) low-throughput in vitro biochemistry, (2) high-throughput cell-based assays, and (3) high-throughput in vitro biochemistry. The first two data types already underpin large, curated repositories; the third stems from emerging methods whose measurements are not yet aggregated into comparable databases but offer distinct advantages.

3.1. Aggregated Low-Throughput Databases. A century of low-throughput in vitro biochemistry has generated equilibrium and rate constants within existing literature, and several efforts have carefully gathered, annotated, and curated these data. Currently, ~800,000 aggregated equilibrium and rate constants describe enzyme kinetics (BRENDA, STRENDa, SABIO-RK)^{42–44}, nucleic acid binding (UniPROBE, ProNAB)^{45,46}, protein–protein interactions (SKEMPI 2.0, PDBbind)^{47,48}, and protein thermostability (ProThermDB, ThermoMutDB)^{49,50}; several initiatives encourage standardized reporting and uploading of data to stored databases (e.g., the STRENDa commission)^{12,13,44}.

Low-throughput in vitro biochemistry characterizes a handful of protein variants at a time under a single set of conditions, a single protein across multiple conditions, or both. This experimental approach creates several persistent problems for data aggregation: (1) experimental conditions (e.g., buffers, salts, etc.) are inconsistently reported⁵¹, (2) measurements may vary due to “hidden” variables (i.e., age of reagents, researcher performing the assay, or other batch effects)^{51–53}, and (3) individual experiments rarely include internal standards or reference materials to allow calibration across assays and validation of measurement quality⁵⁴. A meta-analysis of several papers from leading journals found that none provided the necessary information required to reproduce enzyme activity assays⁵¹. Moreover, measurements performed incorrectly may be properly annotated but inaccurate. In one example, an analysis of 100 papers reporting apparent affinities (K_{DS}) for protein/RNA interactions revealed that only 28/100 measured interactions at equilibrium and at appropriate concentration ranges⁵⁵. In another example, 7/8 published estimates of CRISPR-Cas12 and CRISPR-Cas13 kinetic parameters violated basic physical principles⁵⁶. In each case, a lack of reference measurements within the same assay complicated detection of methodological issues. While ongoing efforts are mitigating variability through further systematization or detailed curation of metadata^{42–50}, these efforts can only standardize what was originally reported and the bulk of preexisting measurements predate rigorous systemization.

3.2. High-Throughput Cellular Measurement Databases. Assays that couple cellular reporters to next-generation sequencing are relatively easy to implement, can achieve extraordinary scale, and can return data that can be modeled to infer equilibrium and rate constants. As a result, ~10 million datapoints detailing nucleic acid binding, protein–protein interactions, enzyme kinetics, and protein stability have been assembled within high-throughput repositories like the Multiplex Assays of Variant Effects (MAVE) database^{57,58}, ProteinGym⁷, and ProtaBank⁵⁹. These data have been used extensively to predict likely pathogenicity for variants of uncertain significance in clinically relevant human proteins and assess the accuracy of protein function prediction algorithms^{8,60}. Within cell-based assays the primary readout is typically a single scalar value (i.e., reporter fluorescence, mating efficiency, or growth rate). While these values are determined in part by the underlying equilibrium and rate constants of variant proteins, they also depend on expression, folding, and cellular contexts. Extracting equilibrium and rate constants from cell-based assays, thus requires both rigorous experimental design and careful biophysical modeling. There are two primary ways to measure constants with cell-based assays: via growth or display.

3.2a. Growth-Based Cellular Assays. Growth-based assays have been applied to binding, stability, and catalysis. For binding, two-hybrid and protein-fragment complementation assays (PCAs) fuse interacting partners to complementary reporter fragments (e.g., split dihydrofolate reductase (DHFR), whose reconstitution rescues growth under methotrexate)^{61,62}. Fitting data from PCAs designed to assess binding and abundance to biophysical models (ddPCA/MoCHI)

made it possible to infer folding and binding free energies (ΔG_{fold} , ΔG_{bind}) for several SH3 and PDZ domain/peptide interactions and K-Ras regulatory interactions^{63–69}. For catalysis, growth measurements can exploit the dependence of cellular fitness on either complementation of an essential enzyme or coupling of a resistance enzyme to antibiotic survival. These approaches have produced sequence-fitness maps for β -lactamases (TEM-1), aminoglycoside kinases (APH), and DHFR^{70–72}; when combined with explicit metabolic-flux models and in-cell substrate titrations, these assays have inferred catalytic parameters ($k_{\text{cat}}/K_{\text{MS}}$), switch-cycle kinetics, and allosteric landscapes for DHFR mutants, GTPases, and plant hormone receptors^{71,73–75}. However, cell-based growth and fitness assays convolve binding and catalysis with abundance and stability. As a result, recovered estimates can be unreliable when mutations promote misfolding, aggregation, or proteostatic burden unrelated to the target reaction (as documented for TEM-1 mutations)^{76,77}; moreover, observed mutational landscapes can vary dramatically with differences in co-expressed folding or quality control proteins^{71,78}. Finally, growth-based assays can only profile activities that can be wired to growth, biasing coverage toward metabolic and resistance enzymes and complementation-compatible interactions.

3.2b. Display-Based Cellular Assays. In display-based assays, each variant is expressed as a fusion to a cell-surface anchor; separating cell populations (e.g., by fluorescence-activated cell sorting, FACS) followed by deep sequencing of sorted pools reports a per-variant biophysical property^{79–82}. The canonical application for these assays is binding, which can be performed as a titration series^{79,83–86}. Incubating displayed variant libraries with increasing concentrations of fluorescent ligand, sorting into occupancy bins based on intensities, and sequencing sorted bins enables reconstruction of a binding isotherm and fitting to yield an absolute affinity (K_{D}) for every variant in parallel (Tite-Seq)⁸². Similar approaches have quantified relative differences in binding free energies across a protease/inhibitor interface and screened binding of candidate antigens against different parental antibodies (MAGMA-seq)⁸⁷; synthetic agglutination (SynAg) reprograms yeast mating so that the mating efficiency of two haploid cells reports the affinity of their displayed partners, enabling library-on-library measurements in high throughput⁸⁸. While most display-based assays use yeast (with proteins displayed as Aga2p fusions)⁷⁹; mammalian display enables production of proteins requiring specific processing pathways (e.g., full-length IgG)⁸⁶. Nevertheless, display-based assays carry challenges arising from surface presentation. Readouts may reflect unwanted interactions with other cell surface proteins and multivalent display can introduce avidity that flattens apparent affinities⁸⁵. Measurements also reflect impacts of mutations on the efficiency with which a variant is expressed, folded, secreted, and displayed at the surface and can be confounded by host-specific post-translational modification⁸⁹.

3.3. High-Throughput In vitro Biochemistry. Emerging high-throughput in vitro biochemical methods measure thousands-to-millions of equilibrium and rate constants in a single experiment outside of cells. A database built from high-throughput in vitro measurements thus offers two main advantages over other databases. First, every variant within a library is measured under identical, well-defined conditions, facilitating highly accurate, internally normalized measurement of relative energetic differences between variants (e.g., $\Delta\Delta G_{\text{bind}}$, $\Delta\Delta G_{\text{fold}}$). Second, the scale of these experiments allows for inclusion of reference standards that flag unreliable measurements prior to data aggregation and enable calibration to yield absolute energetic differences (e.g., ΔG_{bind} , ΔG_{fold}). High-throughput in vitro biochemistry typically tracks individual variants within an experiment via spatial, spectral, or sequence-based multiplexing.

3.3a. Spatial Multiplexing. Spatial multiplexing miniaturizes assay volumes and arrays isolated variants within plates, devices, or droplet arrays. Physically separated reactions proceed in parallel, allowing direct readout of each reaction and systematic titration of reactants (i.e., DNA, RNA, enzymatic substrates, or other proteins) and reaction conditions (i.e., ionic strength, specific ions or molecules, pH, or denaturant) across compartments. Observing each reaction in isolation also makes it possible to follow reaction progress over time to measure kinetics and derive rate constants.

3.3b. Spectral Multiplexing. In spectrally barcoded assays, sequence variants are uniquely associated with and identified by an optical signature (e.g., a ratiometric fluorescent or lanthanide-emission "code" embedded in a bead displaying each variant). In spectral barcoding, the overall throughput is set by the number of resolvable spectral codes (typically 10s to 1000)^{90,91}.

3.3c. Sequence Multiplexing. Tagging variants with unique DNA barcode identifiers enables measurement of variant pools, increasing throughput by resolving variants through next-generation sequencing. These experiments first partition variant pools by properties of interest (e.g., bound *vs.* unbound, folded *vs.* unfolded, reacted *vs.* unreacted). Downstream high read-depth sequencing quantifies the frequency of each variant in each partition, yielding probabilities that reflect differences in free energies. Assays with NGS readouts can profile $>10^5$ variants in a single experiment under identical conditions—nearly two orders of magnitude beyond the throughput of high-throughput spatially multiplexed assays⁹². Nevertheless, extracting physical constants from pooled experiments requires careful modeling and accounting for potential interference between variants.

3.4. Complementarity of Cell-Based and In vitro Assay Formats. Combining data from high-throughput in vitro biochemistry and cell-based assays for the same library provides novel insights not possible from either measurement alone. Apparent differences between in vitro and cell-based measurements can provide critical information about the cellular environment and the degree to which processes like cellular localization or post-translational modification not recapitulated in vitro modulate/regulate function. Moreover, jointly training on abundant cell-based data measurements and smaller, targeted in vitro datasets can connect cellular phenotypes to their physical causes within the relevant context⁴¹, revealing why a variant alters function and more accurately predicting the effects of sequence variation across many proteins.

4. Protein-Ligand Binding.

Protein-ligand binding encompasses an enormous range of interactions that are all governed by the equilibrium and rate constants defined below. Here, we focus on transcription factor (TF)–DNA interactions and protein–protein interactions (PPIs) because: (1) they account for a large fraction of high-throughput in vitro biochemistry assays and (2) both partners are defined by their sequence and easily assembled into large libraries that can be measured at scale. While protein–small-molecule interactions and protein–RNA interactions are compatible with sequence-defined libraries both have been reviewed elsewhere^{93–96}.

The simplest thermodynamic model of protein–ligand binding is a 1:1 bimolecular interaction in which a single protein binds a ligand to form a complex (**Figure 3a**). For such a two-state process at equilibrium, the concentrations of free and bound molecules determine the dissociation constant:

$$\text{Eqn. 10: } K_D = \frac{[P][L]}{[P \cdot L]},$$

where $[P]$ is the concentration of free protein, $[L]$ is the concentration of free ligand (which itself could be a protein), and $[P \cdot L]$ is the concentration of bound complex. K_D represents the concentrations of these species (in units of M) when the free energy of the system is minimized. Experimentally, measuring the exact concentrations of all three species is often intractable. Instead, K_D is typically determined by identifying the concentration of total added ligand at which half of the binding sites are occupied under simplifying experimental conditions (**Figure 3b**), modeled with a binding isotherm based on fraction bound (θ):

$$\text{Eqn. 11: } \theta = \frac{[P \cdot L]}{[P \cdot L] + [P]} = \frac{[L]_{\text{free}}}{K_D + [L]_{\text{free}}} \approx \frac{[L]_{\text{total}}}{K_D + [L]_{\text{total}}}.$$

These simplifying experimental conditions require: (1) that free ligand is approximately equal to total added ligand ($[L]_{\text{total}} \approx [L]$), ensured by adding free ligand in large excess of total protein, and (2) that the fixed protein concentration is well below the K_D (such that binding measurements return binding energy rather than stoichiometry)^{55,97}. Mutations can shift the overall free energy landscape by changing the probability that a given protein–ligand complex is bound at a given concentration, quantified as a change in the energetic difference between the states (or $\Delta\Delta G$) (**Figure 3c,d**).

Measurements of binding kinetics provide additional insights into the mechanisms by which mutations alter affinities (**Figure 3e**). A mutation that alters molecular contacts in the transition state alone slows rates of association (k_{on}) and dissociation (k_{off}) by changing $\Delta G^\ddagger$ without changing ΔG_{bind} . By contrast, mutations that destabilize the bound complex simultaneously reduce ΔG_{bind} and speed k_{off} without changing $\Delta G^\ddagger$; mutations that alter properties of the unbound conformational ensemble impact ΔG_{bind} and k_{on} simultaneously, sometimes revealing states difficult to characterize structurally⁹⁸.

Outside of biochemistry experiments, proteins typically encounter a wide variety of potential ligands. Thus, understanding and predicting binding (and ultimately engineering highly specific interactions) requires quantifying affinities to a target of interest and all possible additional targets a protein may encounter. A given binding landscape should be defined by measuring binding affinity across as many potential targets as possible, which allows quantifying both specificity (the identity of preferred sequences) and selectivity (the energetic difference between a given ligand and other sequences, $\Delta\Delta G$ s)⁹⁸.

4.1. Transcription Factor–DNA Binding. Sequence-specific binding of transcription factor proteins (TFs) to accessible regulatory sequences in DNA regulates gene expression. Unlike equilibrium binding at a single site, which follows the two-state protein/ligand model above (**Figure 3a**), a TF can encounter essentially any stretch of DNA in the genome, each described by its own equilibrium constant. Multiple models have been developed to predict effective binding free energies for a given sequence, from linear mononucleotide models (e.g., additive position weight matrix and energy matrix models)^{99–101} to models that incorporate nucleotide interdependencies and DNA shape effects^{102–107} to deep learning models trained directly on high-throughput binding data^{108–110}. Partition function models then predict apparent affinities for a given stretch of DNA by summing the statistical (Boltzmann) weights of every possible binding configuration. The predicted occupancy of any site is its own weight normalized by the sum of all competing states. These models have successfully predicted observed impacts of low

complexity binding and local sequence context^{101,110–112}. Recently developed interpretable machine learning models can learn complex higher-order dependencies to accurately predict levels of binding at nucleotide resolution, providing in silico ‘oracles’ to further spur understanding^{41,109}.

4.1a. Current High-Throughput In vitro Technologies.

Spatially Multiplexed Assays. Protein-binding microarrays (PBMs) are the earliest and most widely adopted spatially-arrayed assay, with abundant data available within the UniPROBE database^{45,113}. PBMs give an array-based, non-equilibrium intensity readout of a fluorescently labeled TF binding to thousands of user-specified double-stranded DNA sequences. Early studies assessing binding of TFs across structural families to universal "k-mer" DNA libraries characterized TF motif specificities and how mutations altered them^{114–117}. More recently, PBM variants (including genomic-context PBMs) have returned measurements correlated with binding free-energy differences ($\Delta\Delta G$ s)^{107,118–120}.

Microfluidic devices measure surface-bound TFs against soluble DNA at equilibrium, mechanically trapping complexes to retain transient or weak binders through washing (MITOMI, Mechanically Induced Trapping of Molecular Interactions). Early iterations quantified absolute (K_D) and relative ($\Delta\Delta G$) affinities for a single TF against 100s–1000s of printed duplexes (MITOMI, MITOMI 2.0), and demonstrated that in vivo occupancies could be recapitulated by thermodynamic models parameterized with in vitro binding affinities^{121–126}. Successive generations have harnessed devices to recombinantly express and purify TF variants (STAMMP, for Simultaneous Transcription Factor Affinity Measurements via Microfluidic Protein Arrays)^{98,127}; kinetic versions of these assays (k -MITOMI and k -STAMMP) that quantify dissociation rate constants (k_{offS}) have revealed how TFs with similar affinities can bind with substantially different kinetics^{98,112,128}.

A third strategy increases scale without a loss of measurement resolution by repurposing the optics and spatially fixed, sequence-identified clusters of a next-generation sequencer as an *in-situ* array. HiTS-FLIP imaged binding of a fluorescently labeled TF from yeast to extract K_D s for 82,000 DNA sequences on a single flow cell¹²⁹; more recently, the platform quantified K_D s and k_{offS} for the human KLF1 TF¹³⁰.

Sequence Multiplexed Assays. Another family of assays quantifies TF–DNA binding affinities by incubating a purified TF with a library of DNA sequences, physically partitioning bound from unbound molecules, and sequencing these populations to compute relative binding energies. SPEC-seq and BUNDLE-seq separated bound and unbound fractions via electrophoretic mobility shift and first established this approach could recover relative affinities between a single TF and thousands of DNA motif variants for complexes separable by stoichiometry^{131–133}. SMiLE-seq and BET-seq trap TF–DNA complexes on a microfluidic device prior to eluting and deep sequencing; by quantifying the probability that a given variant is in the bound and unbound fractions, BET-seq quantified $>10^6$ $\Delta\Delta G$ s in a single assay and revealed how bases flanking the core motif systematically tune affinity^{105,134,135}. More recently, PADIT-seq replaced physical partitioning with a synthetic genetic circuit that converts binding strength to a transcriptional output, allowing comprehensive measurement of $\Delta\Delta G$ s for all 10-mer sequences across several TFs¹³⁶.

Another family of assays descended from SELEX (systematic evolution of ligands by exponential enrichment), in which a protein iteratively binds and selects high-affinity

binders from a large pool of randomized DNA^{137,138}. Coupling SELEX to a next-generation sequencing readout (SELEX-seq, HT-SELEX) enabled parallelized investigation of many factors in parallel^{139–144}; models that consider sequence enrichment across multiple rounds have been used to extract quantitative binding affinities from these data^{110,145}.

4.1b. Future Directions. Despite substantial progress, coverage of sequence space remains sparse for most TFs and uneven across TF families; kinetic measurements required to predict in vivo transcription from first principles are especially scarce^{146,147}. Further development of platforms capable of quantitative library-on-library measurements will provide critical insights into how changes in either TF or DNA sequence mediate affinity, specificity, and selectivity and rewire transcriptional networks.

4.2. Protein–Protein Interactions. PPIs play essential roles in metabolic pathways, signaling, and the immune response and are the basis of an expanding class of biological therapeutics – antibodies, engineered binders, and other protein drugs. As a result, there is considerable interest in reliably predicting and engineering specific, high-affinity PPIs. Structure-based deep-learning methods can now generate de novo binders for well-structured targets with defined binding sites with variable success rates^{10,148}. Predicting binding affinity and specificity more generally—particularly for the conformationally flexible interfaces that drive most cellular signaling—remains an open frontier, limited by the availability of equilibrium and kinetic constants at scale. This section focuses on high-throughput PPI assays that measure these constants.

4.2a. Current Technologies.

Spatially Multiplexed PPI Assays. Holdup assays equilibrate tagged proteins with immobilized peptides prior to a fast-filtering step (without washing) to separate bound from unbound proteins; subsequent parallelized capillary electrophoresis or mass spectrometry quantifies depleted proteins to return binding affinities^{149–152}. STAMMPING (for Simultaneous Trapping of Affinity Measurements via a Microfluidic Protein-Protein INteraction Generator) leverages MITOMI microfluidic devices to mechanically ‘trap’ soluble partners bound to surface-immobilized bait prior to washing, yielding absolute dissociation constants (K_{Ds}). Initial work quantified affinities between 204 intrinsically disordered human TF activation domains (ADs) and 8 co-activator protein and domains¹⁵³; more recently, ADAPT-M linked yeast surface display-based enrichment to STAMMPING-based affinity measurements for rapid, purification-free quantification of computationally designed binders¹⁵⁴. Another strategy leveraged Illumina sequencing hardware and tethering of proteins to sequenced DNA via ribosome stalling (Prot-MaP) to yield qualitative estimates of binding for high-affinity sequences but did not quantify actual affinities¹⁵⁵.

Spectrally Multiplexed PPI Assays. MRBLE-pep (Microspheres with Ratiometric Barcode Lanthanide Encoding-peptide) synthesizes peptides directly on lanthanide-encoded hydrogel beads engineered to slow association and dissociation kinetics, effectively ‘freezing’ interactions at equilibrium^{90,91}. Incubating beads with titrations of soluble ‘prey’ proteins followed by imaging to quantify occupancies and identify variants can yield 100s of protein/peptide interaction K_{Ds} within a single pooled assay¹⁵⁶ and has been used to quantify ~700 interactions between peptides and human phosphatases^{157–161}. However, these same slowed kinetics prevent measuring protein association or dissociation rates. Beyond the MRBLE system, researchers have demonstrated the ability to quantify K_{Ds} between 12 or 24 interactions between candidate antigens and scFvs produced in cell lysates and captured on fluorescently encoded beads¹⁶².

Sequence Multiplexed PPI Assays. As mentioned in **Section 3.2b**, cell display-based approaches encompass much of high-throughput based assays for PPIs but these assays convolve measured K_D s with cellular processes^{79,85,86}. While phage display is used routinely to display large libraries of peptides and select peptides that bind antibodies or folded protein domains, the low copy number of displayed peptides precludes the fluorescence-based sorting required for TiteSeq. As a result, these assays typically return only information about relative binding specificities. Recently, models trained on multi-round affinity selection data from phage display have yielded predicted K_D s that correlate with experimentally-measured values¹⁶³, suggesting a scalable approach for quantifying protein/peptide affinities.

A limited number of fully in vitro methods have quantified binding constants at scale. High-Throughput Sequencing Kinetics (HTSK) incubated bead-immobilized Bcl-xL with an mRNA display library encoding 20,000 peptide ligands (enriched from a prior screen) and tracked peptide abundance over time via deep sequencing¹⁶⁴. Fitting data acquired during the approach to equilibrium and after washing returned association (k_{on}) and dissociation (k_{off}) constants, enabling direct calculation of K_D s by assuming two-state binding ($K_D = k_{off}/k_{on}$). While powerful, this approach requires ultra-high-affinity binders (pM to low nM K_D s) with very slow off-rates (k_{offs} of 10^{-6} to 10^{-7} /s). More recently, APB-Display (Amplicon/Protein Bead Display), which couples cell-free expression of co-localized protein and encoding DNA on hydrogel beads to a sequencing-based titration (APB-TiteSeq)¹⁶⁵, yielded absolute K_D s for >18,000 FLAG/anti-FLAG interactions in <72 hours; combining a single-concentration measurement with neural network denoising further extended quantitative throughput to 88,000 affinities (K_D s).

4.2b. Future Directions. Protein binder design continues to advance: deep-learning methods now generate de novo-designed minibinders against diverse targets, with greater success rates during each iteration^{10,166–168}. This progress reflects widely available data from enrichment screens and evaluation based on a binary classification of whether a protein binds at all. Designed minibinders are also rigid, which makes it possible to treat them as fixed structures¹⁶⁶.

Predicting and engineering binding interaction affinities, by contrast, remains an unsolved challenge. In silico studies have established that accurate prediction of specific binders will require simultaneously increasing the scale of binding measurements and the diversity of measured sequences³⁹. Therefore, there should be further development of broadly accessible and high-throughput library-on-library technologies that can cover a broad range of orthologous sequences and measure equilibrium and rate constants across several potential binding partners. Finally, learning to predict and engineer the affinities—and the binding networks—of domains means reckoning with their dynamics, because these proteins exist as conformational ensembles that require consideration of multiple states to model effectively.

5. Conformational Ensembles.

5.1. Models for Protein Conformational Heterogeneity.

Proteins are synthesized as linear chains of amino acids and often must fold to function. In the simplest case of global stability, we can model this as a two-state process such that the unfolded (U) and native (N) states are described by the following equations (**Figure 4a**):

$$\text{Eqn. 12: } K_{\text{fold}} = \frac{[N]}{[U]};$$

$$\text{Eqn. 13: } \Delta G_{\text{fold}} = -RT \ln(K_{\text{fold}}).$$

However, the reality is substantially more complex. Every protein samples an ensemble of conformations, and protein function (including binding and catalysis) emerges from this ensemble rather than from any single static structure^{169–171}. Moreover, the conformational transitions relevant to function span a wide range of energies, length scales, and timescales^{170,172}. No single method can observe the full range of all three scales, and the experimental challenge becomes identifying an observable that resolves the scales and states of interest. As stated in **Section 1**, describing the ensemble begins by collapsing it into a small number of states determined by the process of interest, the observables available, and the question being asked. Once states are defined, the measurable quantities are those introduced in **Section 1.4**: the equilibrium constants (K s) that set their relative populations and the rate constants (k s) that set their interconversion. We focus the following discussion on global stability, as it is one of the most fundamental descriptions of conformational heterogeneity.

5.2. Global Stability. The relative populations of the native “folded” state and globally unstructured “unfolded” state define the global stability of a protein (**Figure 4a**). This is the most coarse-grained description, and, by a wide margin, the most abundant measurement related to conformational ensembles. It is also consequential: most proteins and nearly all enzymes must fold to function¹⁷³, unfolded proteins are degraded more readily and are more aggregation-prone^{78,80,174–177}, stability confers mutational robustness required for evolution¹⁷⁸, and destabilization is a common mechanism among disease-associated mutations^{179,180}. This two-state approximation, although a considerable simplification, describes small single-domains well^{181–184}, providing strong priors for understanding the behavior of multi-domain proteins^{185,186}

5.3. Strategies for Measuring Stability. Any measurement of global stability should satisfy several conditions¹⁸⁷. Most importantly, folding must be reversible and the measurement made after relaxation to equilibrium; an irreversible step such as aggregation means the observable no longer reports on the folding equilibrium, while incomplete relaxation means that equilibrium has not been reached. As a final challenge, the equilibrium constant for global folding is often very large: under native conditions, the unfolded state may be too rare to detect directly^{188,189}. Two families of approaches address this: denaturation and native-state exchange.

5.3a. Denaturation. In this approach, a perturbation (typically temperature or a chemical denaturant such as urea or guanidinium chloride) is systematically increased until the unfolded and folded states are at equilibrium and each sufficiently populated to quantify (**Figure 4b**)^{187,190}. The relationship between the free energy of folding and the concentration of chemical denaturant can be conveniently approximated as linear (**Figure 4c**)^{191–193}, allowing for the measurement of an apparent K_{fold} at each condition and an extrapolation to obtain the K_{fold} under native conditions (**Figure 4d**)^{192,193}. The extrapolation requires assuming a number of states (usually two); when that assumption is inappropriate, it yields values with no physical meaning^{194–196}. The use of temperature as a denaturant is complicated by irreversibility: unfolding at high temperature frequently leads to aggregation, and whether equilibrium analysis remains valid depends on how far that irreversible step is separated in temperature from the unfolding transition itself^{181,187,197}. The temperature dependence of stability is also more complex: because the native and unfolded states differ in heat capacity, stability depends non-linearly on temperature^{182,193}.

Because a diverse array of observables can distinguish native from unfolded states, this approach is widely accessible. Traditionally, this involves measuring a spectroscopic signal that reports on secondary or tertiary structure (e.g., circular dichroism or tryptophan fluorescence, respectively);

however, any technique that can distinguish conformational states can be used to quantify populations. Proteolysis is a convenient probe for monitoring denaturation that leverages the high stability of thermophilic proteases (e.g., thermolysin) and the fact that proteases cleave locally unstructured segments that native states rarely present¹⁹⁸. Single-molecule methods resolve individual molecules transitioning between states in real time, directly yielding the populations and interconversion rates for detectable states within the ensemble^{199–201}.

5.3b. Native-State Exchange. These approaches use a reaction that modifies only the unfolded state (**Figure 4e**)^{188,202,203}. For this category of approaches, there are two kinetic regimes that determine whether one is measuring a K_{fold} or a k_{unfold} . (**Figure 4f**). In one regime, the interconversion between the folded and unfolded states is fast relative to the intrinsic rate of the modifying chemistry (commonly referred to as the EX2 regime). As a result, the observed rate of modification reports on the fraction unfolded under native conditions and yields the thermodynamic stability (ΔG_{fold}). In the regime where the modification chemistry is much faster than folding (commonly referred to as the EX1 regime), the rate of modification instead quantifies the unfolding rate (k_{unfold})^{202,203}. Several chemistries have been used in this strategy. Hydrogen-deuterium exchange (HDX) labels backbone amide protons, where the exchange event requires breaking of secondary structure hydrogen bonds and exposure to solvent^{188,202,203} and is typically monitored by nuclear magnetic resonance (NMR) or mass spectrometry (MS)^{203,204}. Thiol exchange involves use of reagents that modify cysteine residues, reporting on whether a given cysteine is buried or exposed to solvent^{205–208}. Similar to its use in denaturation experiments, proteolysis exploits the requirement for extended unstructured segments typically only present in the unfolded (or partially unfolded) states^{92,209–211}. Since this measurement takes place under native conditions, other proteases (such as trypsin and chymotrypsin) can be used^{92,209,210}.

Native-state exchange strategies share a key limitation: assuming the measurement is reporting on global stability requires interpretation with a model that assumes two states. If a modification site (i.e., backbone amide, buried thiol, protease cleavage site) is exposed in a partially folded state more populated than the globally unfolded state, the measurement will quantify the population of that lower energy state (**Figure 4g**)^{188,189,194,211,212}. This can be advantageous, especially for methods like HDX that provide many site-specific measurements per protein, as it reports on states within the ensemble that many other methods are blind to^{188,189,203,205,207}. However, this becomes a particular problem for proteolysis. Proteolysis requires only that a cleavable segment become transiently accessible; because that cleavage is irreversible, the observed rate will be determined by the lowest energy cleavable state^{211,212}. The likelihood that a partially unfolded state is populated increases with protein size and the number of domains²¹³, complicating application of this method to proteins greater than ~100 residues. Compounding this, the site of cleavage is rarely known, so the intrinsic rate cannot be assigned to the segment that unfolded and is instead typically calibrated based on the assumption of a globally unfolded state. Ultimately, if the two-state assumption is not valid in a proteolysis experiment, the state being quantified and the reference required to calculate K_{fold} are undetermined. Fortunately, this problem is diagnosable: because the modification rate can be measured as a function of chemical denaturant, an experiment can establish whether cleavage proceeds through a locally unfolded rather than globally unfolded state (**Figure 4g**)^{188,189,203,211}.

5.4. The State of High-Throughput Measurement.

5.4a. “Brute-Force” Measurements. Aggregation and automation remain useful strategies for assembling datasets, especially for measurements of the conformational ensemble beyond global stability, like those provided by NMR, that resist traditional spatial, spectral, and sequencing-based multiplexing^{172,214,215}. Reanalysis of existing aggregated data can also provide new insights into conformational heterogeneity; re-refining crystallographic data with multiconformer models has recovered occupancy-weighted side-chain heterogeneity across tens of thousands of PDB entries^{216–218}. Large numbers of global stability measurements from the literature have been aggregated within databases like FireProtDB, but the aggregated values were measured under heterogeneous conditions²¹⁹. Automated chemical denaturation of nearly every GB1 point mutant (918 total) monitored by tryptophan fluorescence produced a benchmark set still used to validate new high-throughput stability assays²²⁰.

5.4b. Cell-Based Approaches. Many in-cell assays report variant abundance directly on a scale set by the assay rather than in energetic units, but abundance reflects folding along with expression and turnover^{78,80,174,175}. This can be partially addressed by fitting orthogonal assays jointly and inferring a ΔG_{fold} , as in doubleddeepPCA^{64,66}, which has been applied to a diverse array of systems^{63–65,68,69,221}. Mass spectrometry-based approaches can monitor proteins from complex mixtures through a denaturation experiment at scale^{222,223}: Chemical denaturation at lysate scale (SPROX) relies on irreversible methionine oxidation^{223,224}, while Thermal Proteome Profiling (TPP) and Limited Proteolysis (LiPs) return apparent melting temperatures across entire proteomes^{225,226}. However, this gain in throughput comes with costs that differ by method. SPROX returns transition midpoints whose conversion to free energies assumes a two-state transition that likely does not hold for a proteome comprised of many large and multidomain proteins¹⁹⁶, while TPP relies on protein aggregation and measurement of the remaining soluble fraction such that the method does not claim to measure equilibrium properties or constants^{225,227}.

5.4c. Spatially Multiplexed In vitro Approaches. Spatially multiplexed microfluidic platforms trade throughput for real-time observation of many independent reactions under precisely controlled solution conditions. SPARKfold leverages valved microfluidic devices for parallel expression and native-state proteolysis of protein variants under EX1 conditions to quantify unfolding rate constants (k_{unfold}) across many sequences in parallel²²⁸. The initial demonstration quantified unfolding rates for 31 DHFR variants with up to 78 replicates each. A related microfluidic approach quantifies rates of catalysis for 100s of surface-immobilized enzymes in the presence of increasing urea concentrations¹⁷. Applying this approach to the human SHP2 protein tyrosine phosphatase yielded measurable denaturant melts; additional catalysis and ligand binding experiments revealed conformational heterogeneity among SHP2 variants, demonstrating that combining kinetic measurements with thermodynamic modeling can report on conformational ensembles.

5.4d. Sequence-Multiplexed In vitro Approaches. Among in vitro approaches, sequence-multiplexed proteolysis provides the clearest case of true physical constants at high throughput. Displaying tens of thousands of designed miniproteins on the surface of yeast and exposing the library to varying concentrations of trypsin or chymotrypsin was used to extract global stabilities at scale²¹⁰. cDNA-display extended this to a fully in vitro format by linking each protein covalently to its own cDNA (allowing for a true in vitro experiment) and fitting the resulting proteolysis rates to a thermodynamic model. This yielded $\sim 776,000$ absolute ΔG_{fold} values measured under identical conditions⁹² and was recently extended to millions of small domains within the MGnify database, providing a large dataset of quantitative global stabilities used to

train predictive models²⁰⁹. However, these approaches convert the rate of proteolytic cleavage into a ΔG_{fold} of folding under a two-state assumption²¹¹. While this is an appropriate assumption for the small domains in those experiments, there are obstacles towards applying this assay towards larger proteins. As larger proteins are more likely to populate intermediates that expose hydrophobic segments that will also be cleaved by protease^{188,189,195,211}, naively interpreting those proteolysis rates will quantify the difference between the native state and that partially folded state rather than global unfolding (**Figure 4g**).

5.5. Future Directions. The overwhelming majority of global stability measurements come from systems chosen for reversible two-state behavior because choosing such proteins makes the measurement practical, not because reversible two-state folding is a general or even common property of proteins²²⁹. The proteins with these convenient properties are not representative²³⁰. Nearly all global stability measurements in existence come from datasets restricted to single domain proteins less than 80 residues in length^{92,209,219}, but the median human protein is greater than 350 residues and often comprises multiple domains^{186,231}. Two-state reversible proteins are also much more likely to be soluble, monomeric, cytoplasmic, and free of disulfide bonds, post-translational modifications, and cofactors²³⁰. While the ensembles of more complex systems can still be studied, doing so is more experimentally challenging. Multi-state systems can be measured by denaturation, but separating the intrinsic stability of one unit from its coupling to its neighbors is difficult (as decades of work on repeat proteins has shown¹⁸⁵) and resolving more complex local unfolding has required HDX and NMR^{188,189,194,203}. The consequences of this skew are most visible in what can currently be predicted and validated.

Predicting the simplest constants describing a conformational ensemble— $\Delta\Delta G_{\text{fold}}$, and absolute ΔG_{fold} for small single domains under defined conditions—is increasingly accessible^{209,232} and enabled by the arrival of large, internally consistent datasets^{92,209}. Generative models have begun to extend this: BioEmu was trained on both a clustered version of the PDB and large volumes of molecular dynamics simulation to generate equilibrium conformational ensembles; subsequent model fine-tuning on high-throughput global stability data from cDNA-display proteolysis enabled prediction of folding free energies^{92,233}. Further progress on predicting diverse scales of conformational heterogeneity will require experimental data that do not currently exist at scale.

Ensemble-aware predictive models that target populations and dynamics rather than static structures will require populations and interconversion rates of states within the folded ensemble, not only the global folding stability. Progress is underway: HDX-MS applied across designed protein energy landscapes recently recovered distributions of local opening free energies rather than a global unfolding equilibrium across thousands of small domains²³⁴. This is the first large-scale dataset to begin resolving local stability and dynamics for many sequences in parallel, although it returns a distribution per sequence rather than site-resolved values. The frontier is now extending this progress to include varied solution conditions, more diverse protein systems, and broader scales of conformational heterogeneity.

6. Enzyme Catalysis.

6.1. The Model of Enzyme Catalysis. Enzyme catalysis is highly complicated but can often be modeled using a simple three-state system (**Figure 5a**). In the Michaelis-Menten model of enzyme kinetics, free enzyme (E) binds a substrate (S) to form a "Michaelis complex" E•S, which is then converted to free enzyme and product (E+P) (**Figure 5b**). Catalysis is a kinetic description of molecular behavior and is rarely examined under systems that are fully at

equilibrium, focused more on quantifying $\Delta G^\ddagger$ s (the heights of transition-state barriers) than ΔG s (see **Section 1.4b**)^{20,235}. In enzyme kinetics, these are derived by measuring reaction velocities and rate constants (k s). Measured for different variants, this process can identify specific interactions that selectively stabilize transition states over ground states to lower reaction barriers, the defining feature of catalysis.

A typical experiment titrates substrate and measures the initial velocity of product formation to generate data that can be fit to the Michaelis-Menten model, yielding two parameters:

$$\text{Eqn. 14: } \frac{v_0}{[E]} = \frac{k_{\text{cat}}[S]}{K_M + [S]},$$

where v_0 is the observed reaction velocity, $[E]$ is the total enzyme concentration, k_{cat} is the maximum rate of turnover (per molecule of enzyme), and K_M is the Michaelis constant.

The similarity of this equation to **Eqn. 11** suggests an equivalence of the parameter K_M to K_D . Indeed, K_M approximates the binding affinity of E to S (K_D) when a quasi-equilibrium can be established between the E+S and E•S states, which is true when the rate constant from substrate dissociation (k_r) is much faster than k_{cat} . However, if k_{cat} is comparable to (or much faster than) k_r , then equilibrium cannot be meaningfully established because the concentration of E•S is not determined solely by the probabilities given by K_D for substrate binding. The Briggs-Haldane assumption assumes instead that the concentration of E•S is constant over time but not necessarily in equilibrium with E+S, yielding an enzyme system that is at “steady state”—having constant flux through the system—providing a constant rate of product formation over time. Steady state kinetics, the predominant form of enzyme measurements, therefore precludes the strict interpretation of the parameter K_M as a K_D . Although k_{cat} and K_M are the commonly solved parameters, enzymologists interpret k_{cat} and k_{cat}/K_M as the two fundamental kinetic parameters and infer K_M from these^{236,237}.

The Michaelis-Menten model does not explicitly include a state for the product-bound enzyme, E•P (where S has been converted to P within the enzyme active site but not released; **Figure 5c–d**). As a result, the inverse of the measured parameter k_{cat} corresponds to the time it takes the enzyme to do the chemistry and release the product (**Figure 5e**). If one of these steps is much slower than the other, then: (1) k_{cat} can be approximated as the rate constant of that step (**Figure 5f**), and (2) the three-state model is an appropriate representation of the physical process. (Note, however, that the enzyme-bound state may correspond to E•P rather than E•S, again complicating the physical interpretation of K_M). Because standard steady-state rates can rarely distinguish between a three-state model and a four-state model, the simpler approximation is used. Finally, while it may feel intuitive to assume that chemistry is rate limiting, this is likely the exception rather than the rule in natural enzymes²³⁸, especially when using reactive fluorogenic substrates²³⁹. When chemistry is not rate-limiting, mutational effects do not report on the chemical reaction mechanism.

6.2. Scalable Enzyme Assays. Enzyme behavior is amenable to measurement, as many analytical methods can detect the products of an enzymatic reaction and signal-to-noise ratios are amplified for catalytic outputs. As a result, many databases of catalytic constants exist and have been widely used^{7,240,241}. However, k_{cat} and k_{cat}/K_M , even when assuming a simplified model, must be derived from many measurements (**Figure 5g**), all of which must be carefully controlled. Non-spatially multiplexed assays common to high-throughput approaches (i.e., pooled assays) are

complicated by the fact that released product diffuses away from the enzyme that produces it, making it difficult to link variant genotype with phenotype. Moreover, it is often difficult or labor-intensive to quantify enzyme concentration, which is essential to calculate k_{cat} and k_{cat}/K_M ²⁴².

Here, we highlight new experimental methods for quantifying enzyme catalysis at scale, limiting our focus to those that can assay many (>100) different protein sequences.

6.2a. “Brute-Force” Spatial Multiplexing. The requirement for time-resolved measurements over multiple substrate (and potentially enzyme) concentrations makes multi-well plates invaluable for measuring enzyme parameters, but experiments are generally still limited to only a few different protein variants per plate. A notable effort scaled this approach to perform hundreds of highly standardized plate-based assays to determine k_{cat} , k_{cat}/K_M , and thermal stability metrics for 100 and 129 purified single-mutants of a glycoside hydrolase (BglB)^{243,244}. This has since been expanded to >800 curated variants of the same enzyme in the Design2Data (D2D) database²⁴⁵, returning true k_{cat} and k_{cat}/K_M values for most variants and including critical data on expression, stability, and computationally predicted metrics.

6.2b. Sequencing-Based Multiplexing. Enzymes that do not perform multiple turnovers or release product (and therefore are arguably not “catalysts”) can be easily profiled via pooled assays, as products remain physically associated with the variants that generated them. If they are tagged with DNA (or are themselves nucleic acid-modifying enzymes), their sequences can be enriched in downstream selection assays. Such assays have been applied to self-aminoacylating RNA enzymes (ribozymes) via a pooled multiplexed assay in which self-aminoacylation with a biotin-tagged substrate allowed capture of sequences according to their second-order (k_{cat}/K_M -like, units of /M /s) rates of catalysis (k -Seq)²⁴⁶. If the activity can be tied directly to modification of a piece of DNA that is linked to the genotype, sequencing the number of mutations in DNA—as opposed to the number of DNA molecules—can be used to assay enzyme activities, though these have not yet been used to determine k_{cat} and k_{cat}/K_M values²⁴⁷.

Researchers frequently deploy pooled cell growth assays under conditions in which growth rates are proportional to variant activities, making it possible to track variant abundance over time as a proxy for enzyme activity²⁴⁸. This can be done at massive scale^{249,250} but gaining insights into enzyme catalysis requires deconvolving functional effects from altered catalysis, expression, stability, or other processes. While not measuring k_{cat} and k_{cat}/K_M , some studies have accounted for one or more of these variables by either focusing on substrate selectivity²⁵¹, measuring global stability in parallel²⁵⁰, or systematically varying substrate concentration to obtain K_M -like values²⁴².

Combining large-scale cellular measurements with a set of targeted in vitro measurements for the same constructs can allow “calibration” of cell-based phenotypic data, yielding estimations of k_{cat}/K_M values^{74,75} and even the inhibition constants (K_i s) of inhibitors from growth rates⁷¹. Recent efforts to quantitatively model catalytic parameters from growth-based experimental measurements use approaches to systematically manipulate cellular substrate concentrations and other variables, enabling estimation of k_{cat}/K_M values for thousands of enzyme variants of DHFR²⁵². As mentioned in **Section 3.2**, a common theme of growth-based enzyme assays is that protein quality control plays a major role in determining fitness effects^{76,78}.

6.2c. Microfluidics-Based Spatial Multiplexing. Multiple technologies scale down assay volumes to determine kinetic parameters via high-throughput measurements across many substrate and inhibitor concentrations (e.g., microfluidic chips²⁵³, droplets^{254–257}, or sequencing²⁵⁸); however, most of these require labor-intensive protein purification that do not directly scale the throughput of variant libraries. To alleviate this bottleneck, HT-MEK (High-Throughput Microfluidic Enzyme Kinetics)^{16,17,239,259,260} recombinantly expresses, purifies, and iteratively assays spatially multiplexed variants within thousands of nanoliter reaction chambers in parallel. The first demonstration of HT-MEK shows the power of this approach for generating physical data: this study acquired >670,000 reaction progress curves for 1,036 variants of the alkaline phosphatase superfamily member PafA, yielding >5,000 kinetic and thermodynamic constants (k_{cat} , $k_{\text{cat}}/K_{\text{M}}$, K_{M} , and K_{is}) across multiple substrates and inhibitors²³⁹. These data distinguished mutational effects on misfolding and different steps of the reaction (see **Figure 5e**). Subsequent work has since: examined the effects of transition-state analogs (TSAs) by collecting ~2,000 K_{is} ¹⁶; exhaustively biochemically profiled human allelic variants of the SHP2 protein tyrosine phosphatase by measuring >300,000 progress curves, improving clinical predictions and updating the mechanism of action of established inhibitors¹⁷; generated a sequence–catalysis landscape for 193 adenylate kinase orthologs²⁵⁹; and physically modeled how protease catalysis impacts viral fitness for >400 viral protease variants²⁶⁰.

While HT-MEK dramatically scales the ability to iteratively profile purified enzyme variants, the devices are technically challenging to fabricate, require custom pneumatic control equipment²⁶¹ and a fluorescent assay, and tend to adsorb small, hydrophobic molecules. DA-MEK (Droplet Array Microfluidic Enzyme Kinetics) leverages wettability-patterned superhydrophilic/omniphobic surfaces for high-throughput enzyme expression and measurement of catalytic constants²⁶² in an open water-in-air format.

6.3. Future Directions. Measurements of enzyme activity have already increased our understanding of enzyme evolution, function, and even catalytic mechanisms. But given the complexity of the physical determinants of enzyme catalysis—true even of the simplest cases shown in **Figure 5**—it is reasonable to ask: will scaling steady-state kinetic measurements be enough to learn generalizable rules of enzyme catalysis? Physically grounded measurements of protein variants should report on how mutations impact the energies of specific states and barriers associated with individual steps occurring on the enzyme. Both k_{cat} and $k_{\text{cat}}/K_{\text{M}}$ are composite kinetic parameters that arise from multiple underlying rate constants for discrete, physical events along the reaction coordinate. Pre-steady-state kinetics are invaluable for revealing rate constants for individual steps but currently do not scale to large libraries²³⁹. Modeling, understanding, and predicting enzyme catalysis will benefit from deliberate choices of enzyme systems that are amenable to physical interpretation using scalable methods. This includes ones that: (i) can be modeled using Michaelis-Menten kinetics (i.e., rapid equilibrium binding of substrate, rate-limiting chemistry), drawing equivalence between K_{M} and binding energies; (ii) allow measurement of effects on specific steps (i.e., within more complex reaction schemes); and/or (iii) investigate reactions that can also be modeled with computational approaches (i.e., quantum mechanical/molecular mechanical (QM/MM) modeling of small proteins with no/uncomplicated cofactors). Accounting for these complexities will help realize models capable of predicting more complex enzymatic processes like multi-step reactions, allosteric regulation, and promiscuous reactivity.

7. Conclusions and Outlook

Accurate protein structure prediction was made possible by the availability of >250,000 structures of >45,000 unique proteins now held in the Protein Data Bank (PDB), accumulated since its establishment in 1971. As advances in computational models and hardware accelerate, we must now focus on assembling a similar repository of data to realize an era of accurate protein function prediction. As detailed here, emerging technologies can now quantify equilibrium and rate constants for diverse protein functions in high throughput; simultaneously, DNA synthesis costs are decreasing rapidly, and emerging technologies can faithfully assemble DNA fragments to yield genes encoding larger protein variants.

Existing databases—and even individual high-throughput datasets—already contain several times more equilibrium and rate constants than the PDB has structures, yet achieving even a modest level of generalizable protein function prediction does not yet appear within reach. Ultimately, how much data each prediction task will require is itself unknown, but a more nuanced comparison to structure prediction is informative. The information content in a single structure is dense, explicitly representing tens of thousands of atomic interactions. In contrast, a single equilibrium or rate constant only implicitly represents how those atomic interactions impact the conformational ensemble and how the ensemble changes with environmental conditions. Each individual high-throughput dataset of equilibrium and rate constants likely contains comparable information content to that of a single structure: tens of thousands of internally consistent, physically meaningful data points. The PDB relied on distributed deposition by thousands of laboratories over decades and was effective because of the shared formats and standards that rendered independently collected structures interpretable in aggregate. Emerging high-throughput methods described here provide a path to generating information-dense datasets for many systems and functions, but achieving the scale necessary for protein function prediction will require the same combination of distributed measurement and shared standards.

We suggest that realizing the ability to quantitatively and accurately predict diverse protein function from sequence will require establishing a ‘PDB of function’: a community repository for shared, standardized measurements of protein function, and a framework for the community practices needed to advance the shared goal of predicting protein function. The data within this community repository should satisfy several requirements. First, all data must be standardized, well-annotated, open-source, and easily accessible. Second, all data must describe function using the language of equilibrium and rate constants, which ultimately report on free energy differences. As much as possible, more ‘raw’ forms of data should be preserved such that they can be revisited and remodeled; in all cases, data should be accompanied by all code required to reproduce models and fitted constants. Third, data collection must increase coverage: across conditions, across functions, and across proteins that are currently beyond the reach of existing assays. And finally, a shared panel of reference materials should be measured whenever a new assay is established and alongside each new library so that independently generated datasets can be more readily validated and combined.

The broader community should also engage and interact in several specific ways. Experimentalists and computational researchers must work in tandem to design maximally efficient experiments; as experiments progress, these close collaborations will be crucial for mechanistically interpreting results and developing new hypotheses to be tested. Acquired data should be used to host ‘blinded’ competitions, like the Critical Assessment of Structure Prediction (CASP)^{263–265}, with explicit community-defined criteria for success; as experiments

increase in scale while decreasing in the time required, these competitions should happen with increasing frequency. Synergistic efforts should quantify functional variation for the same libraries in vitro and in cells, simultaneously minimizing costs and uniquely enabling the quantitative comparisons that allow for the discovery of new biological mechanisms. Finally, the community should prioritize the development of low-cost, easily translatable assays and reference materials to enable the crowdsourced measurement of these essential data that will be required to achieve the scale needed to realize generalizable protein function prediction.

Disclosure Statement

The authors are not aware of any affiliations, memberships, funding, or financial holding that might be perceived as affecting the objectivity of this review.

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Figures and Captions

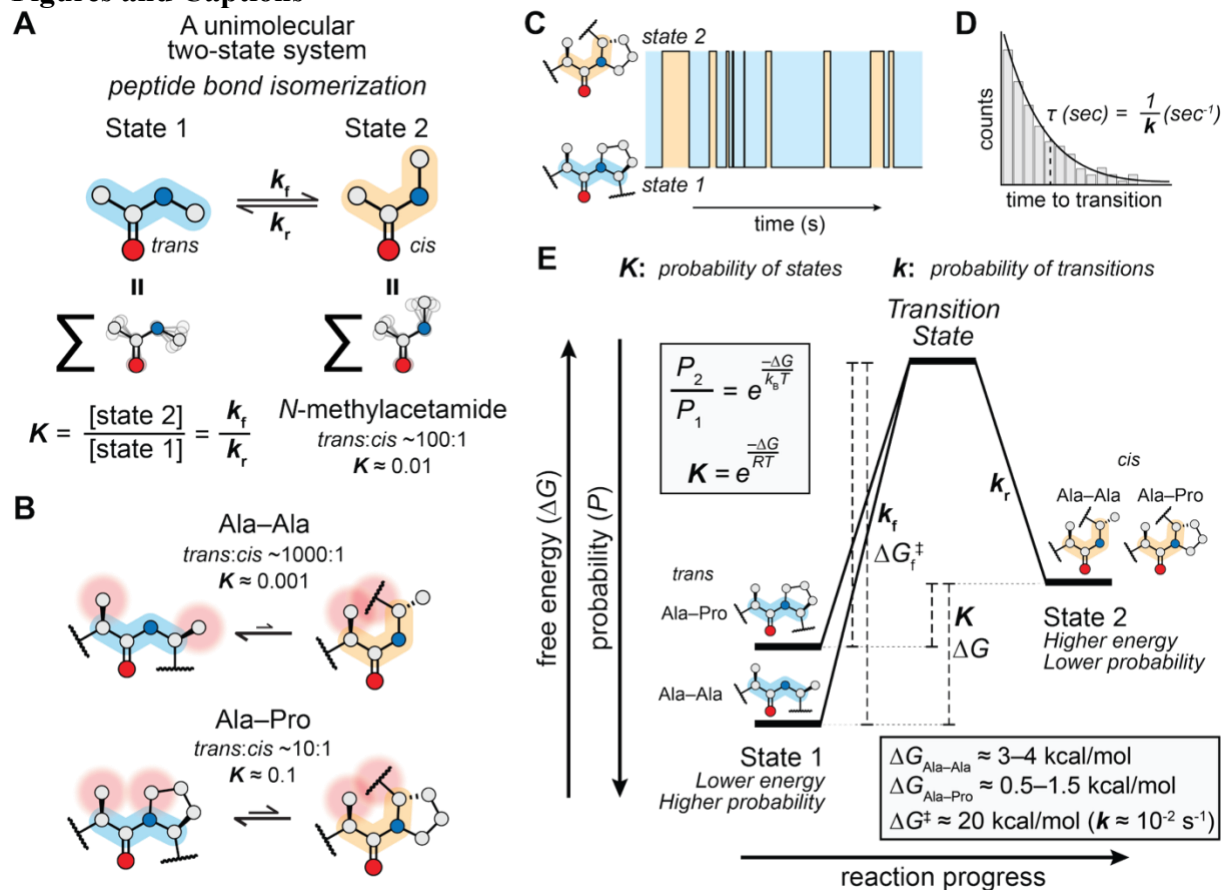


Figure 1. Describing (bio)molecular behavior with physical constants. (a) Rotation of a peptide bond transitioning between two states: *trans* and *cis*. These two macrostates are fully distinguishable but are ensembles comprised of underlying indistinguishable microstates. For *N*-methylacetamide, the *trans* state is favored ~100:1 at equilibrium, giving an equilibrium constant (K) for the reaction of *trans* to *cis* of ~0.01. (b) Steric overlap in the *cis* state for an Ala-Ala dipeptide increases the *trans:cis* ratio ($K \approx 0.001$); the irregular structure of proline in Ala-Pro reduces the *trans:cis* ratio ($K \approx 0.1$). (c) Schematic of a single-molecule trace; the total time spent in each state quantifies the equilibrium constant. (d) Histogram of dwell times within a state and exponential fit yielding a mean time to transition (the inverse of the rate constant for that transition). (e) A free energy diagram with free energy (ΔG) on the y-axis and the reaction progress on the x-axis, which here represents the rotation of the peptide bond. Energies are normalized to the higher-energy *cis* states. The height of the barrier in a given direction represents $\Delta G^\ddagger$, which depends exponentially on the rate constant for isomerization (k) via the Eyring-Polanyi equation.

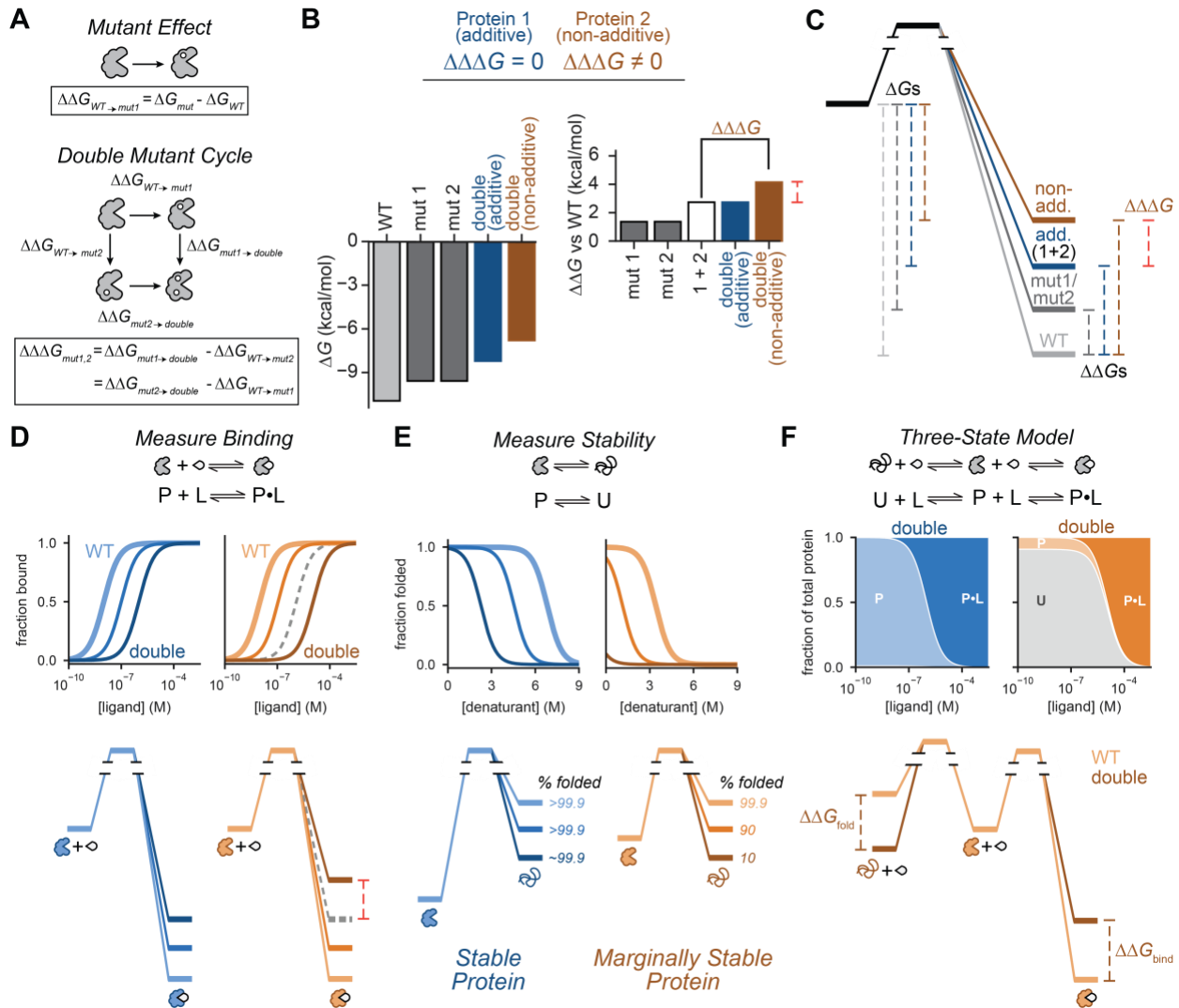


Figure 2. Understanding protein function with free energy measurements. (a) A double mutant cycle. Measuring the free energy change (ΔG) of a process for multiple protein variants—a wild type (WT), two variants with single mutations (mut_1 , mut_2), and a variant with both mutations (double)—and then calculating the difference between pairs of proteins results in $\Delta\Delta G$ s corresponding to mutational effects. $\Delta\Delta\Delta G$ is the difference between the effect of a mutation measured in the background of another mutation versus on its own. (b) Example of two protein scaffolds with the same apparent ΔG for WT and where both single mutants have the same effect. For one protein, the mutational effects are additive (Protein 1, blue, $\Delta\Delta\Delta G = 0$), while they are non-additive in the other (Protein 2, orange, $\Delta\Delta\Delta G \neq 0$). (c) The free energy diagram for each sequence is overlaid, highlighting the same ΔG s, $\Delta\Delta G$ s, and $\Delta\Delta\Delta G$ from (b). (d) A double mutant cycle experiment where apparent ligand binding is measured for the two types of proteins described in (b). (top) Ligand binding titration experiment (see **Section 4**). The measured binding of the double mutant of Protein 2 is weaker than would be expected from additivity (gray line). (bottom) Free energy diagrams for the ligand binding experiment. (e) The same double mutant cycle applied to stability. (top) Denaturant melt experiment (see **Section 5**). When the folding stability of each variant is measured, the $\Delta\Delta G$ s are the same, but one scaffold is stable (blue) and the other is marginally stable (orange) such that the double mutant is mostly

unfolded. (*bottom*) Free energy diagrams for the denaturant melt experiment. (*f*) A three-state model resolves apparent non-additivity. (*top*) The populations of unfolded (U), folded (P), and bound (P•L) states as a function of ligand concentration. (*bottom*) Three-state free energy diagrams for the WT and double mutant variants of Protein 2. When both folding and binding are accounted for the cause of the apparent coupling between mutations (apparent binding $\Delta\Delta\Delta G \neq 0$) is revealed to originate from the large population of an unfolded state.

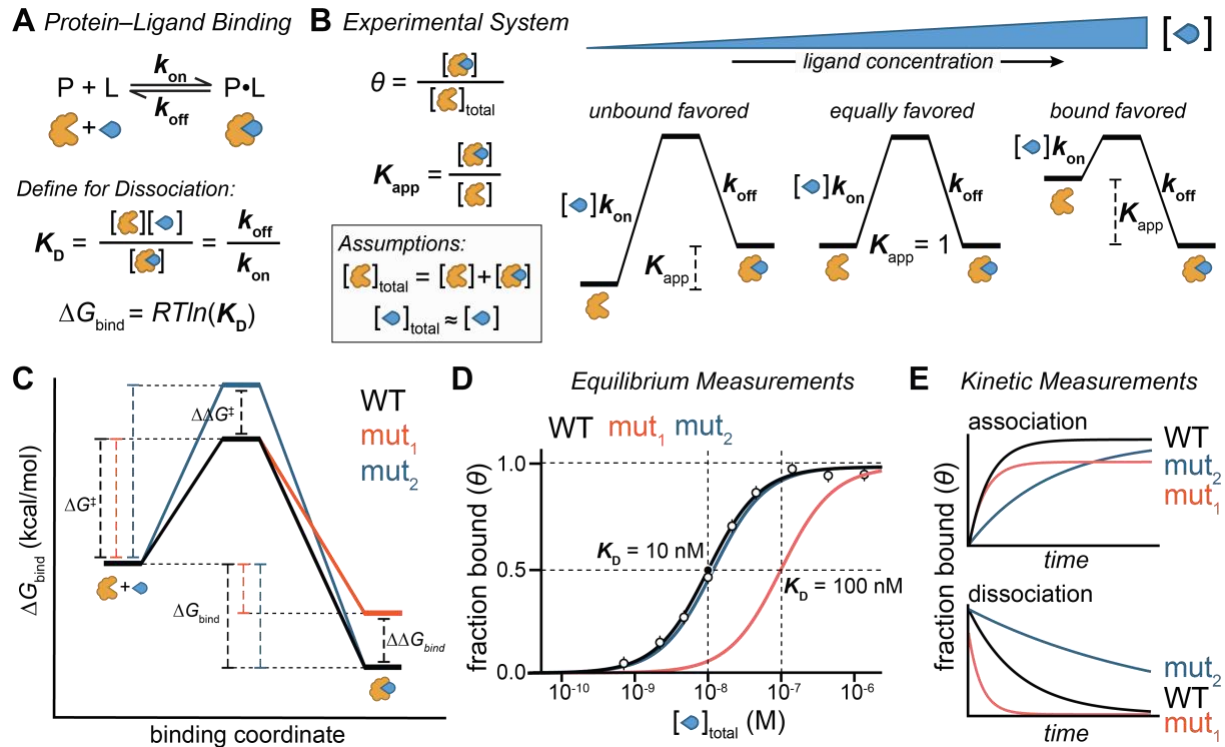


Figure 3. Measuring protein–ligand binding. (a) The model of binding. The equilibrium constant is typically framed in the direction of dissociation (K_D), describing a bound complex $[P \cdot L]$ converting to free protein $[P]$ and free ligand $[L]$. The dissociation constant is defined by the equilibrium concentrations of free and bound species, or equivalently by the ratio of rate constants, and is inversely related to the free energy of binding, ΔG_{bind} . (b) Experimentally, the K_D is commonly determined from the amount of ligand required to reach a fraction bound (θ) of 0.5 (or an equal probability of free and bound species, K_{app}) under simplifying assumptions, namely conservation of mass and that total ligand added is equal to free ligand at equilibrium. (c) Quantitative free-energy diagram for WT and two mutants illustrating the distinct energetic effects of a mutations: mut₁ destabilizes the bound state, while mut₂ raises the transition barrier without changing ΔG_{bind} . (d) Representative equilibrium binding isotherm. Fraction bound is measured as a function of ligand concentration under the assumptions given in (b); K_D corresponds to the concentration at half-maximal occupancy when $[P] \ll K_D$. Curves show WT (black) and two mutants: mut₁ (red) binds relatively weakly, whereas mut₂ (blue) has the same affinity. (e) Association and dissociation kinetic measurements of the binding isotherm for WT and two mutants. Association measures on-rate (k_{on}) at a given high ligand concentration; dissociation measures the intrinsic off-rate (k_{off}), which is independent of ligand concentration.

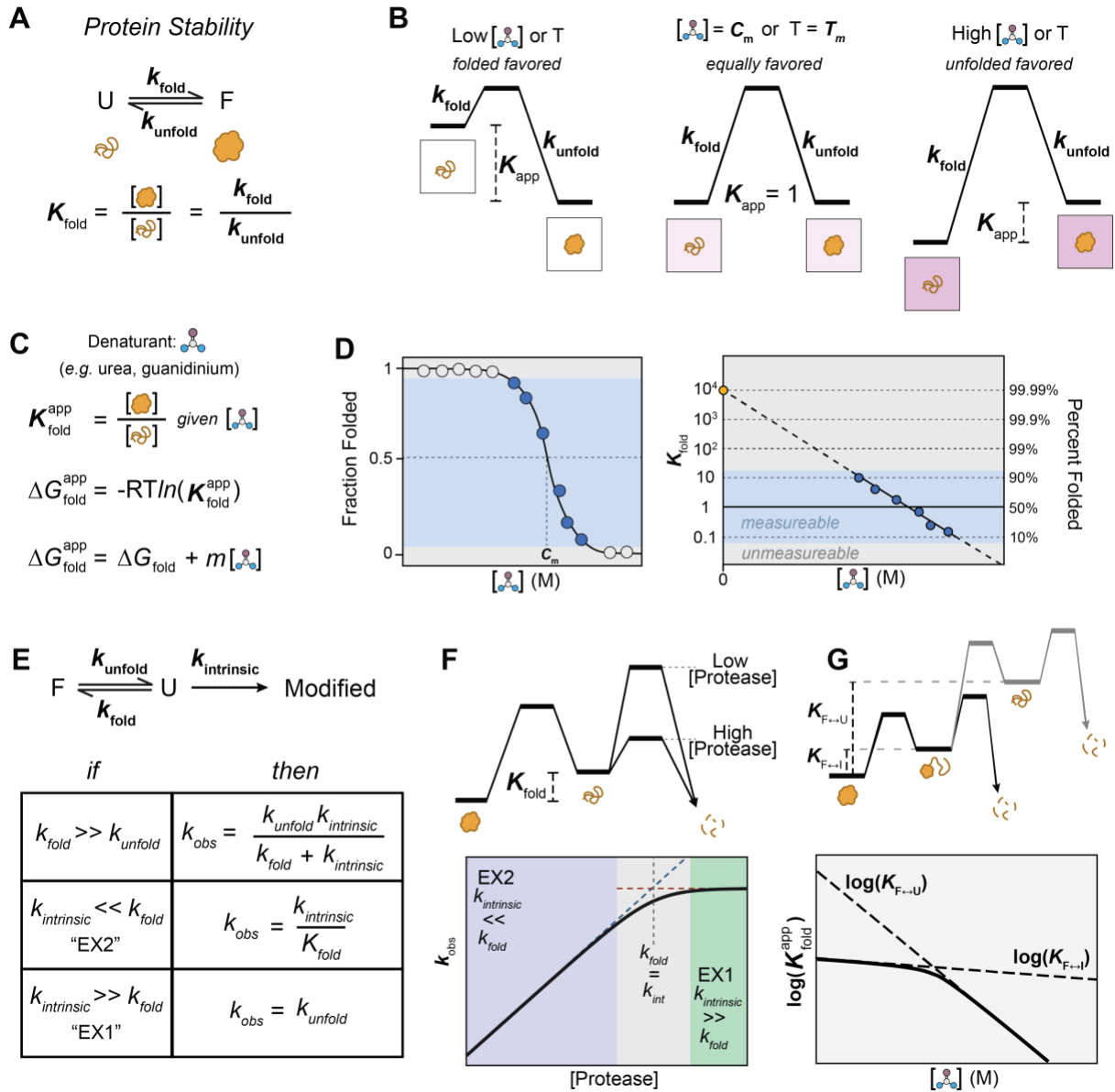


Figure 4. Measuring global stability. (a) A two-state model for folding. (b) Free energy diagram of a folding reaction at increasing denaturant. For chemical denaturant, the free energy of both states is equal at the concentration midpoint or “ C_m ”, and for thermal denaturation at the temperature midpoint or “ T_m ”. (c) Shown are the fraction of folded protein (left) and K_{fold} (right) as a function of chemical denaturant. The logarithm of the folding equilibrium constant is linearly related to the concentration of chemical denaturant. (d) Schematic showing fraction folded as a function of the concentration of chemical denaturant illustrating the expected sigmoidal transition and model fit to extrapolate to zero denaturant conditions and obtain the K_{fold} in native conditions. (e) A model for native-state exchange methods, where a reaction specifically labels the unfolded state. Under conditions where folding is favored ($k_{\text{fold}} \gg k_{\text{unfold}}$), the observed rate of exchange can report on either K_{fold} (EX2) or k_{unfold} (EX1). (f) Modulating the intrinsic rate of modification (by changing the concentration of protease, in native-state proteolysis) can switch the kinetic regime between EX1 and EX2. (g) If a partially unfolded state

is present at a higher population than the globally unfolded state, then the observed rate of proteolysis in the EX2 regime reports on the population of the partially unfolded state. This three-state regime can be diagnosed by performing experiment under varying concentrations of denaturant.

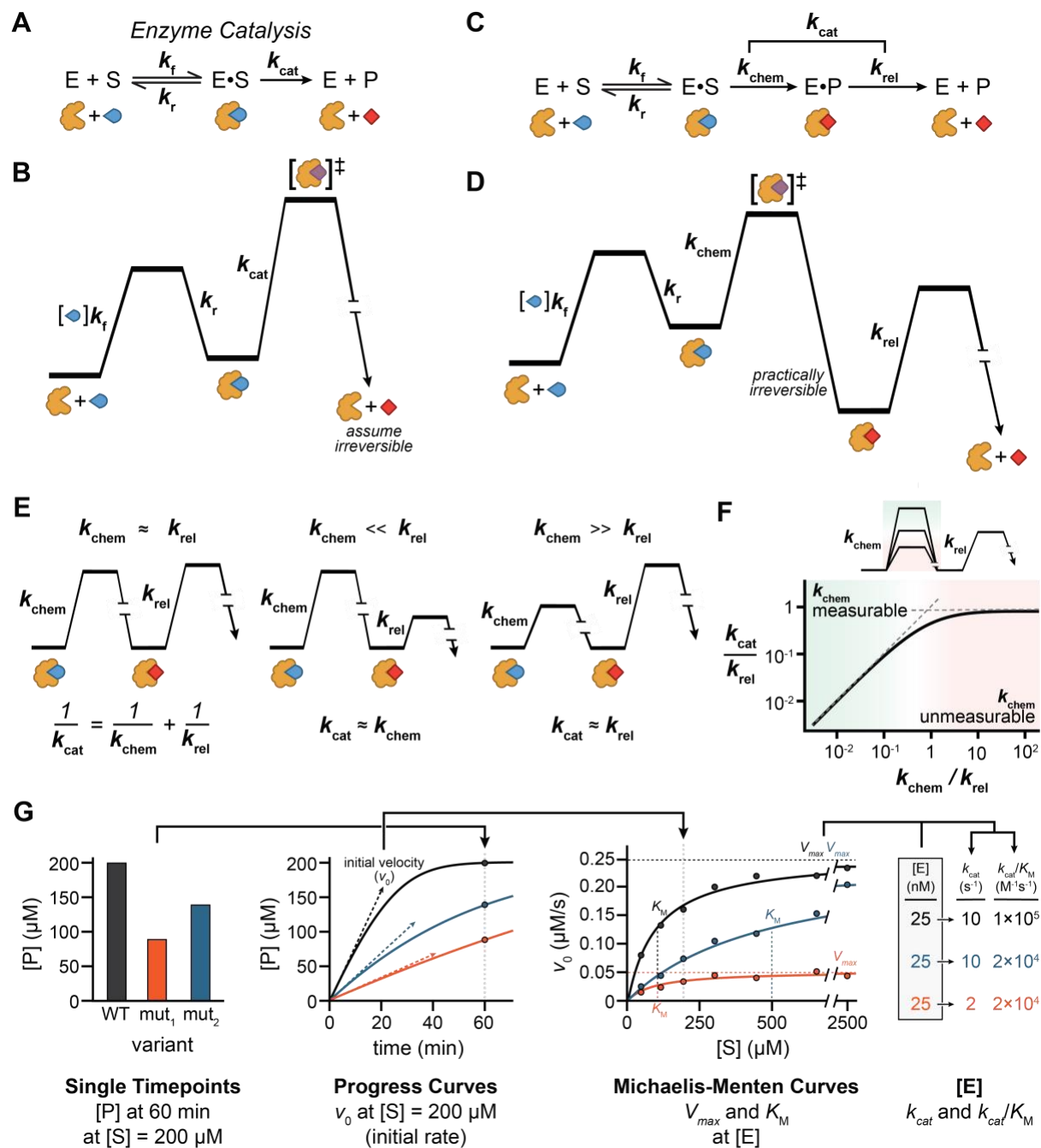


Figure 5. Measuring enzyme catalysis. (a) The Michaelis-Menten model of enzyme catalysis. (b) Free energy diagram for the Michaelis-Menten model, with various substrate concentrations and irreversible product formation. (c) The minimal physical model of enzyme catalysis includes a product-bound state, and k_{cat} reports on both chemistry and product release. (d) Free energy diagram for the four-state model. (e) Interpretation of k_{cat} depends on the relative heights of the barriers for chemistry and product release. (f) The relationship between the ratio of a desired intrinsic rate (here, k_{chem}) and a second, potentially rate-limiting process (here, the rate k_{rel}) compared to the ratio of the observable rate (here, k_{cat}) and that second process. Measurement of the desired rate is increasingly inaccurate for ratios near and above 1 (e.g., $k_{chem} = k_{rel}$). (g)

Building Michaelis-Menten curves. Single timepoint measurements determine product formation of a set interval. The extent of product formation is described by progress curves at the given substrate concentration, from which we can obtain initial velocities (product formation over time). Normalizing for enzyme concentration and measuring many concentrations of substrate, we determine the kinetic parameters k_{cat} , K_{M} , and $k_{\text{cat}}/K_{\text{M}}$.